Sampling techniques

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Sampling methods

Probability sampling method (Random)

The best way to ensure that a sample will lead to reliable and valid inferences is to use probability samples, in which the probability of being included in the sample is known for each subject in the population. The four commonly used probability sampling methods are:

- Simple random sampling;
- Systematic sampling;
- Stratified sampling;
- Cluster sampling.

Simple random sampling

A **simple random sample** (lottery method) is one in which every subject has an equal **probability of being selected** for the study. The recommended way to select a simple random sample is to use a table of random numbers or a computer-generated list of random numbers.
Simple random sample is the simplest form of probability sampling. To select a simple random sample you need to make a numbered list of all the units in the population from which you want to draw a sample or use an already existing one (sampling frame). Then;

* Decide on the size of the sample;
* Select the required number of sampling units, using a lottery method or a table of random numbers.

Systematic random sampling

- It is an alternative to simple random sampling that is useful in some cases.
- The items or individuals of the population are arranged in some order. A random starting point is selected (by lottery) and then every k^{th} member of the population is selected.
- The k is determined by dividing the total number of items in the sampling frame by the desired sample size.

For example, if there are $N = 1000$ stores along Fifth avenue and we want to select $n = 100$ stores in the sample, $k = \frac{N}{n} = 10$. We shuffle only the first k, and select one, say 4. Now on we systematically select stores by adding $k, 2k, 3k, 4k$ etc to 4. So a systematic sample will have store number $4, 14, 24, 34, 44, 54$ etc.

Stratified random sampling

- A **stratified random sample** is one in which the population is first divided in to relevant **strata** (subgroups), and a random sample is then selected from each stratum proportionally.
- Characteristics used to stratify should be related to the measurement of interest, in which case stratified random sampling is the most efficient.
- A population is first divided into subgroups, called strata, and a sample is selected from each stratum. (e.g., 70% males, 30% females).
- If a sample of 10 is selected, (n=10) 70% of $n = 7$, so select 7 males and 3 females. In general, N=population size, N_1=stratum 1(female), N_2= stratum 2 (males), n=sample size desired.
- Sample should have $\dfrac{N_1}{N}n$ individuals from stratum 1 and so on.

Cluster random sampling

- It is a survey procedure in which the sampling units consist of a group of elements known as a **cluster**.
- Information on clusters is used in making inferences about a population.
- The scheme of selection of these clusters could be simple random sampling or systematic sampling.
- Clusters are usually formed by grouping units which can be conveniently observed together.
- These could either be mutually exclusive or overlapping.
- In practice, however, it is more convenient to use clusters that are non-overlapping.
- In cluster sampling, the first task is to specify the appropriate clusters.

Major Advantages of Cluster Sampling

The three major advantages of using cluster sampling in some surveys are:

1. The sampling frame from which the list of individual elements can be obtained may not be available or may not be easily obtained, but that of the clusters may be available and can be obtained easily.
2. It is more expensive to construct a list of elements in a population compared to a list of clusters. Greater costs may be incurred in taking a sample of elements as compared to clusters of elements.
3. The use of cluster sampling usually requires less personnel and takes a shorter time to obtain results

Non random sampling

Non-probability samples are those in which the probability that a subject selected is unknown.

Convenience sampling

It is exactly what the name says. That is one chooses a sample that is convenient. Examples;

1. When checking oranges packed in boxes, only oranges from the top layer are inspected,
2. Examining plants that are easily accessible, for example, found near a road.
3. Sampling patients coming to a specific clinic on a specific day. That is if the clinic is in the morning you may miss children that are at school.

Quota sampling

It is often used for opinion surveys. That is each interviewer is told to collect opinions from a specified number of items.

1. Select quota based on known characteristics of the population.
2. For example, age breakdown of a population Sample would be proportional to the size of each age segment.
3. You do not know whether the individuals that you select from each characteristic are representative of that characteristic.
4. Systematic bias can result.